

Inductor (Kinect)

$$V_L = L \frac{di_L}{dt}$$

$$\bar{i}_L = \frac{1}{L} \int V_L dt$$

更自然的因果 causality

Capacitor (Potential)

$$V_C = \frac{1}{C} \int i_C dt$$

Natural Causality

$$\bar{i}_C = C \frac{dV_C}{dt}$$

Resistor (Dissipater)

$$V_R = \bar{i}_R \cdot R$$

$$\bar{i}_R = \frac{V_R}{R}$$

State Space

$$\dot{x} = Ax + Bu \quad \text{State Equation}$$

$$y = Cx + Du \quad \text{Output Equation}$$

x : state variable (系统内部动态描述的最少变量)

u : input variable (外部施加到系统的信号)

y : output variable (被测/受控输出)

A : state matrix (System property, character 状态自身的演化)

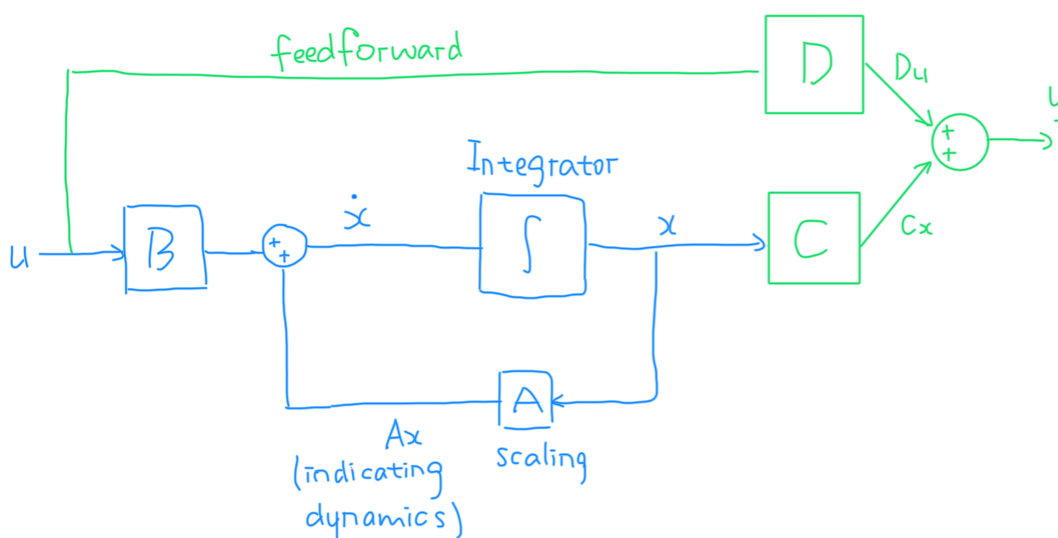
B : input matrix (Scaling input variable)

C : output matrix (状态映射到输出)

D : direct transfer matrix (输入是否直接影响输出)

State Equation:

Output Equation:

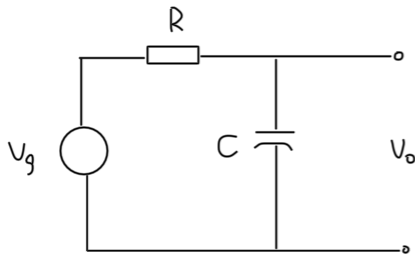


• 所有 state variable 都应是

integrator 的 output

- state variable 具有记忆性
会记住过去的演化
- 代表了动态特性

eg.



① Identify energy storage elements

✖ Order of system == 储能元件数量 (例中为一阶系统)

✖ 阶数 == 最小独立 状态变量 个数

② Identify variables that will be used for modeling

System Response 取决于 1) Input given to system (u)

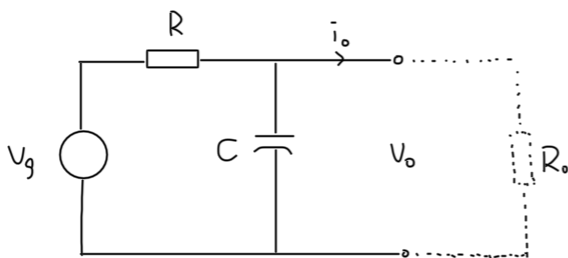
2) Integrator outputs values (state variable x)

Identify all u and x . (例中 $V_c = \frac{1}{C} \int i_c dt$, 所以 V_c 是 x , V_g 是 u)

③ Starting with dynamic elements obtain differential equations

✖ 阶数 == 微分方程 个数

$$V_c = \frac{1}{C} \int i_c dt$$



$$\left[\overset{u}{V_g}, \overset{x}{V_c}, \overbrace{R, C, R_o}^A \right]$$

$$V_c = \frac{1}{C} \int i_c dt = \frac{1}{C} \int (i_R - i_o) dt = \frac{1}{C} \int \left(\frac{V_g - V_c}{R} - \frac{V_c}{R_o} \right) dt$$

$$\therefore \frac{dV_c}{dt} = \frac{V_g - V_c}{CR} - \frac{V_c}{CR_o}$$

④ Segregate into form $\dot{x} = \underset{\text{state}}{Ax} + \underset{\text{input}}{Bu}$

$$\dot{x} = Ax + Bu$$

$\Updownarrow$

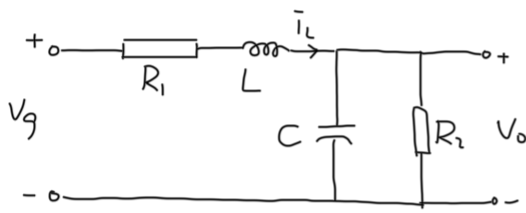
$$\dot{V}_c = \left(-\frac{1}{CR} - \frac{1}{CR_o} \right) V_c + \frac{1}{RC} V_g \quad (\text{state equation})$$

$$y = Cx + Du$$

$$\hat{V}_o = 1 \cdot V_c + 0 \cdot V_g$$

$$V_o = 1 \cdot V_c + 0 \cdot V_g \quad (\text{output equation})$$

Second Order System: (2 differential equations & 2 state variables)



variable list:

$$u = [V_g] \quad \text{input vector}$$

$$x = \begin{bmatrix} i_L \\ V_c \end{bmatrix} \quad \text{state vector}$$

parameters: R_1 , L , C and R_2

$$L: L \frac{di_L}{dt} = V_g - V_{R_1} - V_c$$

$$L \frac{di_L}{dt} = V_g - i_L R_1 - V_c$$

$$\frac{di_L}{dt} = \underbrace{-\frac{R_1}{L}(i_L) - \frac{1}{L}(V_c)}_{Ax} + \underbrace{\frac{1}{L}V_g}_{Bu} \quad (\text{Differential Equation for } L)$$

$$C: C \frac{dV_c}{dt} = i_L - \frac{V_c}{R_2}$$

$$\frac{dV_c}{dt} = \underbrace{\frac{1}{C}(i_L) - \frac{1}{R_2 C}(V_c)}_{Ax} + \underbrace{0 \cdot V_g}_{Bu}$$

$$\begin{cases} \frac{di_L}{dt} = -\frac{R_1}{L}(i_L) - \frac{1}{L}(V_c) + \frac{1}{L}V_g \\ \frac{dV_c}{dt} = \frac{1}{C}(i_L) - \frac{1}{R_2 C}(V_c) + 0 \cdot V_g \end{cases}$$

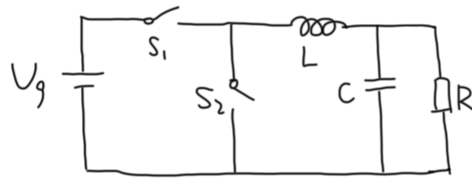
$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_c \end{bmatrix} = \begin{bmatrix} -\frac{R_1}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{R_2 C} \end{bmatrix} \begin{bmatrix} i_L \\ V_c \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [V_g] \quad \text{state equation}$$

$$\dot{x} = A x + B u$$

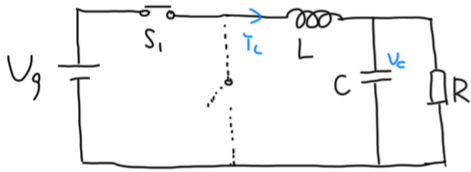
$$V_o = [0 \quad 1] \begin{bmatrix} i_L \\ V_c \end{bmatrix} + [0][V_g] \quad \text{output equation}$$

$$y = C x + D u$$

Buck DC-DC Converter (Discrete)



DT (S₁ ON & S₂ OFF)



$$L \frac{di_L}{dt} = V_g - V_c$$

$$C \frac{dV_c}{dt} = i_L - \frac{V_c}{R}$$

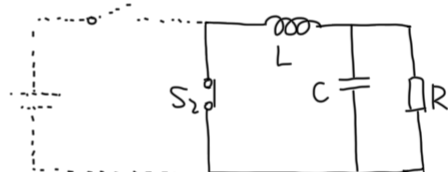
$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_c \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ V_c \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [V_g]$$

$$\dot{x} = A_1 x + B_1 u$$

$$V_o = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ V_c \end{bmatrix}$$

$$y = C_1 x$$

(1-D)T (S₁ OFF & S₂ ON)



$$L \frac{di_L}{dt} = 0 - V_c$$

$$C \frac{dV_c}{dt} = i_L - \frac{V_c}{R}$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_c \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ V_c \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} [V_g]$$

$$\dot{x} = A_2 x + B_2 u$$

$$V_o = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ V_c \end{bmatrix}$$

$$y = C_2 x$$

Large Signal Mode during DT:

$$\begin{cases} \dot{x} = A_1 x + B_1 u \\ y = C_1 x \end{cases}$$

Large Signal Mode during (1-D)T

$$\begin{cases} \dot{x} = A_2 x + B_2 u \\ y = C_2 x \end{cases}$$

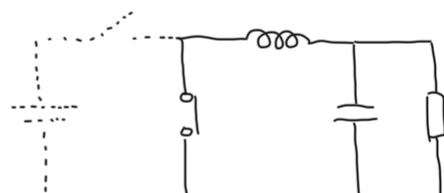
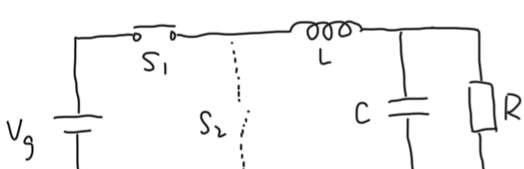
State Averaging

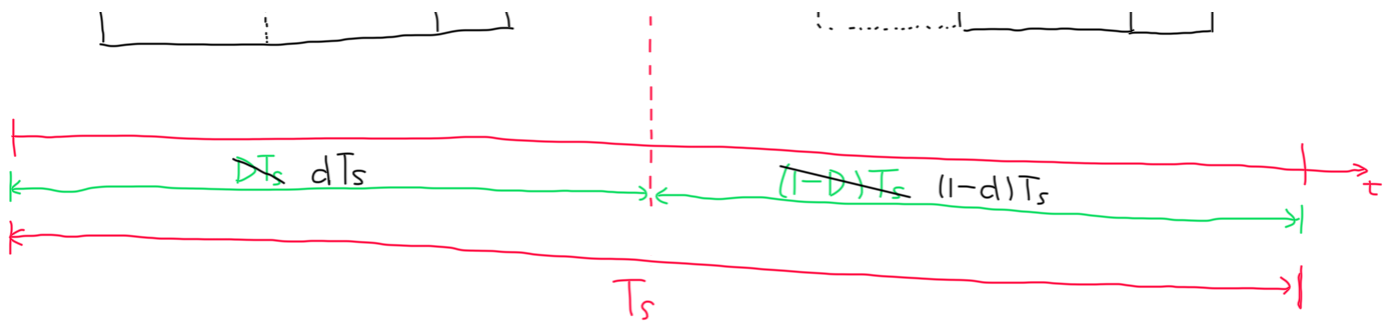
*D

*(1-D)

$$x_{avg} = x_{dT} \cdot D + x_{(1-d)T} (1-D) \text{ (Averaging of Large Single)}$$

Circuit Averaging Method:





$$\begin{bmatrix} \dot{i}_L \\ \dot{V}_C \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ V_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [V_g] \quad \begin{bmatrix} \dot{i}_L \\ \dot{V}_C \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ V_C \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} [V_g]$$

$$V_o = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ V_C \end{bmatrix}$$

$$V_o = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ V_C \end{bmatrix}$$

$$u = U + \hat{u}, \text{ wherein } \frac{\hat{u}}{U} \ll 1$$

$$d = D + \hat{d}, \quad \frac{\hat{d}}{D} \ll 1$$

$$x = X + \hat{x}, \quad \frac{\hat{x}}{X} \ll 1$$

$$y = Y + \hat{y}, \quad \frac{\hat{y}}{Y} \ll 1$$

Large Signal Steady State Small Signal

$$\begin{cases} \dot{x} = A_1 x + B_1 u \\ y = C_1 x + D_1 u \end{cases}$$

$$\begin{cases} \dot{x} = A_2 x + B_2 u \\ y = C_2 x + D_2 u \end{cases}$$

d

$(1-d)$

$$\begin{cases} \dot{x} = A x + B u \\ y = C x + D u \end{cases} \quad \text{Averaged Large Signal Model}$$

$$\begin{cases} A = A_1 d + A_2 (1-d) \\ B = B_1 d + B_2 (1-d) \\ C = C_1 d + C_2 (1-d) \\ D = D_1 d + D_2 (1-d) \end{cases}$$

$$\dot{x} = A x + B u$$

$$(X + \hat{x}) = A (X + \hat{x}) + B (U + \hat{u})$$

$$(X + \hat{x}) = [A_1 d + A_2 (1-d)] \cdot (X + \hat{x}) + [B_1 d + B_2 (1-d)] \cdot (U + \hat{u})$$

$$\textcircled{1} \text{ For } X + \hat{x}, \quad \frac{\hat{x}}{X} \ll 1$$

any term with $(\hat{x} \cdot \hat{d}) \Rightarrow$ neglect

$$\textcircled{2} \text{ Steady State part } X \text{ 相当稳定 on } \frac{dX}{dt} = 0 \quad (i.e., \dot{X} = 0)$$

Since $\lambda = 0$ is not an eigenvalue, $\frac{d}{dt} X = 0$ ($X = AX + BU = 0$)

$$\dot{\hat{x}} = A\hat{x} + B\hat{u} + [(A_1 - A_2)X + (B_1 - B_2)U] \hat{d}$$

Control input

$$y = Cx + Du$$

$$Y + \hat{y} = [C_1 d + C_2 (1-d)](X + \hat{x}) + [D_1 d + D_2 (1-d)](U + \hat{u})$$

$$\hat{y} = C\hat{x} + D\hat{u} + [(C_1 - C_2)X + (D_1 - D_2)U]\hat{a}$$

Averaged Model of Buck Converter

$$A = A_1 d + A_2 (1-d)$$

$$= \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} d + \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} (1-d)$$

$$= \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix}$$

$$B = B_1 d + B_2 (1-d)$$

$$= \begin{bmatrix} \frac{1}{L} \\ 1 \\ 0 \end{bmatrix} d + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} (1-d)$$

$$= \begin{bmatrix} \frac{d}{L} \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{D}{L} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \dot{\hat{i}_L} \\ \dot{\hat{v}_c} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} \hat{i}_L \\ \hat{v}_c \end{bmatrix} + \begin{bmatrix} \frac{D}{L} \\ 0 \end{bmatrix} [\hat{v}_g] + \begin{bmatrix} \frac{D}{L} \\ 0 \end{bmatrix} [\hat{d}] \quad \text{Small Signal Model}$$

$$\begin{bmatrix} \dot{\hat{i}_L} \\ \dot{\hat{v}_c} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} \hat{i}_L \\ \hat{v}_c \end{bmatrix} + \begin{bmatrix} \frac{D}{L} & \frac{DV_g}{L} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{v}_g \\ \hat{d} \end{bmatrix}$$

